Opening Questions

Sports

- Who is the best quarterback of all time?
- Are baseball pitchers throwing harder than ever?
- How much more do male soccer players earn than females?
- How common is it for former Olympic athletes to become coaches?
- How much does an extra inch of height help a basketball player?

Pop Culture

- What percentage of people have seen the movie that won last year's Best Picture Award?
- Who tends to be more popular: bands or solo singers?
- Are younger actors paid more than older actors?
- Are movies with female leads as profitable as movies with male leads?
- Does winning a Grammy increase sales?

Politics

- Is "Stop and Frisk" a racist policy?
- Do Republican politicians tend to come from different states than Democratic ones?
- Do people in countries that have universal healthcare live longer than people in countries that don't?
- Was press coverage slanted for or against a particular candidate?

Education

- Do small schools perform better than large ones?
- Which has a stronger correlation with student achievement: race or wealth?
- Do bilingual classes result in better outcomes for ESL/ELL students?

What Questions Can You Answer with the Given Data?

The following is a dataset of a bicycle rider's training rides.

date	miles	time (w/stops)	weather	average speed	max speed
04/10/2018	10	44	"cloudy"	13	30
05/30/2018	15	66	"sunny"	13.5	22
06/12/2018	12	61	"rainy"	11.2	25
07/04/2018	24	103	"sunny"	14	26
07/12/2018	24	120	"windy"	12.5	26

1) Decide whether each questions below *can* or *cannot* be answered with the given data and circle your selection.

2) In the space provided below, explain *how* you could answer the question using the data or *why you cannot* answer the question.

Question	Answered by this dataset?
How many miles did the cyclist ride June 12th?	*Yes _No*
What tire pressure produces the highest average speed?	*Yes* _No
What is the average time it takes this cyclist to ride 1 mi?	*Yes _No*
Does this cyclist ride slower when it is rainy?	*Yes* _No
Does this cyclist ride faster when they are late to an appointment?	*Yes* _No
How many miles has the cyclist ridden in total as part of their training?	*Yes _No*

Are there any questions that you could find the answers to more than one way?

Domain and Range

My Definition

Facts & Characteristics

Domain

Examples

Non-examples

My Definition

Facts & Characteristics

Range

Examples

Non-examples

Contracts

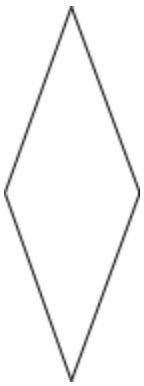
Contracts tell us how to use a function. For example: `ellipse :: (Number, Number, String, String) -> Image` tells us that the name of the function is `ellipse`, it takes four inputs (two Numbers and two Strings), and it evaluates to an `Image`. From the contract, we know `ellipse(50, 100, "solid", "teal")` will evaluate to an `Image`.

Name		Domain		Range
`# triangle`	`::`	`Number, String, String`	`->`	`Image`
`triangle(80, "solid", "darkgreen")`				
`# star`	`::`		`->`	
`# circle`	`::`		`->`	
`# square`	`::`		`->`	
`# rectangle`	`::`		`->`	
`# rhombus`	`::`		`->`	
`# ellipse`	`::`		`->`	
`# text`	`::`		`->`	
`# regular-polygon`	`::`		`->`	
`#right-triangle`	`::`		`->`	

Using Contracts (continued)

Use the contracts to write expressions to generate images similar to those pictured.

`rhombus :: Number, Number, String, String -> Image`

	
Write an expression for a square (rotated) using `rhombus`!	
What variable changes with the first Number?	
What variable changes with the second Number?	

Triangle Contracts (SAS & ASA)

Type each expression (left) below into the editor and match it to the image it creates (right).

Expression			Image
`triangle-sas(120, 45, 70, "solid", "black")`	1	A	
`triangle-sas(120, 90, 70, "solid", "black")`	2	B	
`triangle-sas(120, 135, 70, "solid", "black")`	3	C	
`triangle-sas(70, 135, 120, "solid", "black")`	4	D	

Think about how you would describe each of the arguments that `triangle-sas` takes in to someone who'd never used the function before and annotate the contract below using descriptive variable names.

```
triangle-sas :: (
    ____________________________________________ :: Number,
    ____________________________________________ :: Number,
    ____________________________________________ :: Number,
    ____________________________________________ :: String,
    ____________________________________________ :: String
) -> Image
```

Add examples of each of the triangle functions we've explored to your contracts page.

If you have time, experiment with the `triangle-asa` function.

```
triangle-asa :: (
    left-angle :: Number,
    left-side :: Number,
    bottom-angle :: Number,
    style :: String
    color :: String
) -> Image
```

Star Polygon

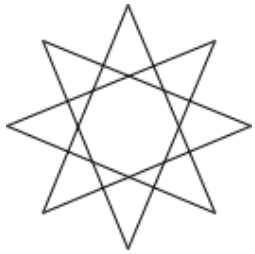
```
star-polygon :: (

    side-length :: Number,
    points-on-polygon :: Number,
    polygon-points-to-skip-between-star-points :: Number,
    shading-style  :: String,
    color :: String

) -> Image
```

Using the detailed contract above, write expressions to create each image below.

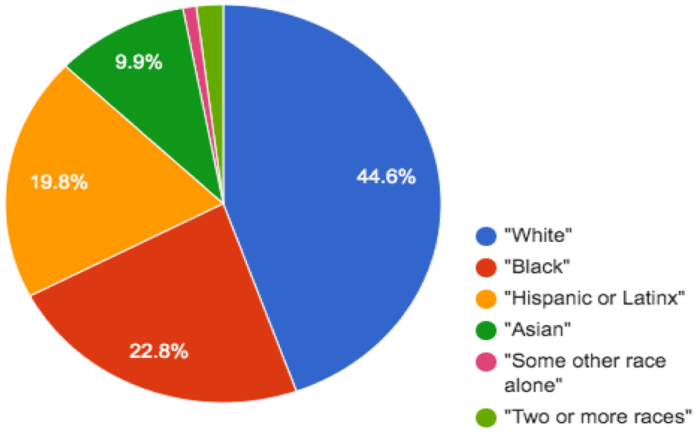
Then make two more star polygons of your choosing. Sketch them and write expressions to generate them.

	_____
	_____
	_____
	_____

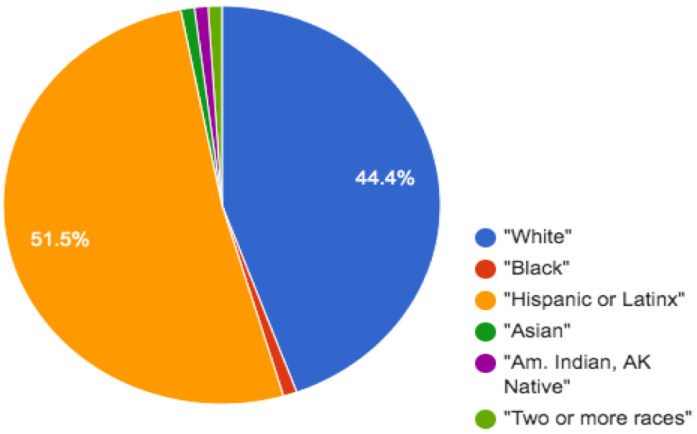
Pie Chart - Notice and Wonder

What do you notice and wonder about the pie charts below?

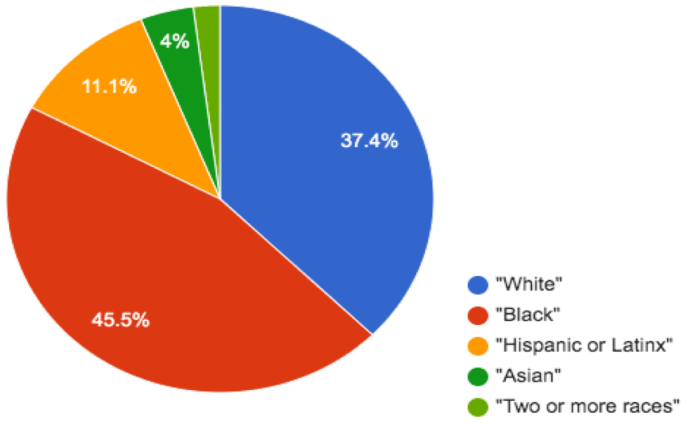
Boston School District, MA



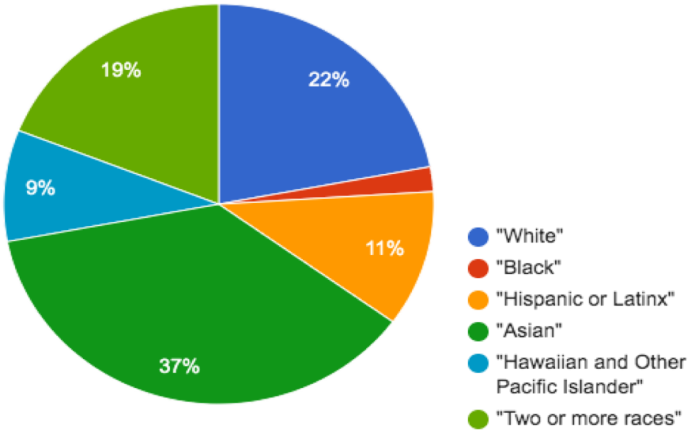
Santa Fe Public Schools, NM



Washington, D.C. Public Schools



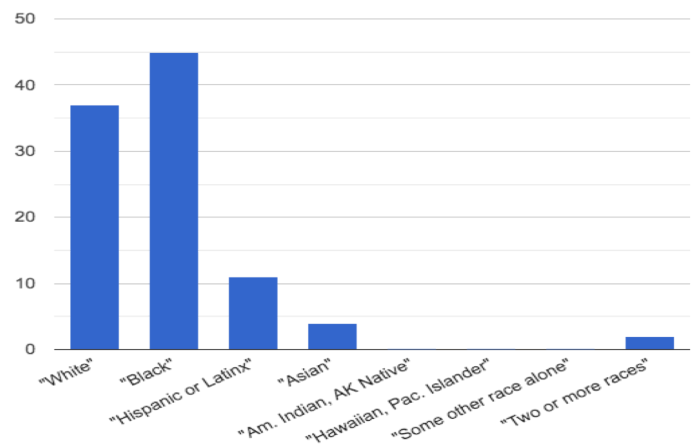
Hawaii DOE, HI



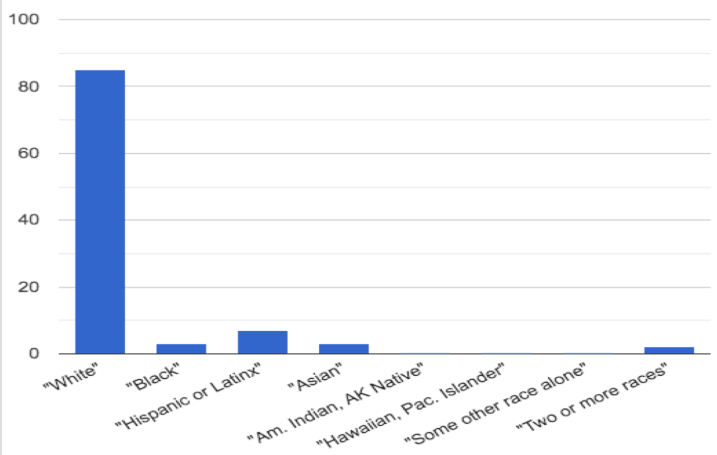
What do you Notice?	What do you Wonder?

Bar Chart - Notice and Wonder

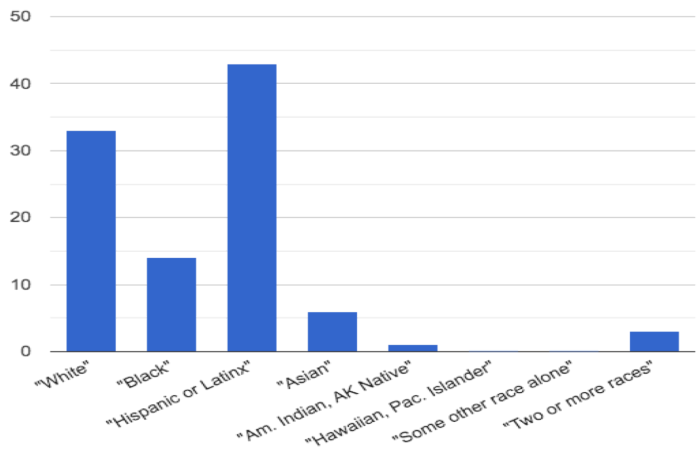
Washington, D.C. Public Schools



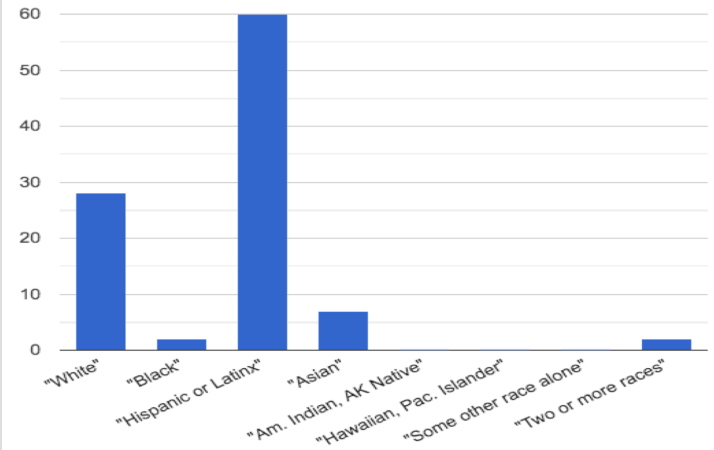
West Warwick Schools, RI



Providence Schools, RI



Oxnard Union HS District, CA



What do you Notice?	What do you Wonder?

Question Types: Animals

A subset of the whole Animals Dataset is shown in the table below.

name	species	sex	age	fixed	legs	pounds	weeks
Sasha	cat	female	1	false	4	6.5	3
Sunflower	cat	female	5	true	4	8.1	6
Felix	cat	male	16	true	4	9.2	5
Sheba	cat	female	7	true	4	8.4	6
Billie	snail	hermaphrodite	0.5	false	0	0.1	3
Snowcone	cat	female	2	true	4	6.5	5
Wade	cat	male	1	false	4	3.2	1
Hercules	cat	male	3	false	4	13.4	2
Toggle	dog	female	3	true	4	48	1

Using this table - or the full dataset - write three questions of each type below.

	Question	Type
1		Lookup
2		Lookup
3		Lookup
4		Arithmetic
5		Arithmetic
6		Arithmetic
7		Statistical
8		Statistical
9		Statistical

Predictions from Samples

1) In the Definitions Area of the Expanded Animals Starter File, define the following samples:

```
tiny-sample = random-rows(more-animals, 10)
small-sample = random-rows(more-animals, 20)
medium-sample = random-rows(more-animals, 40)
large-sample = random-rows(more-animals, 80)
```

2) Click "Run" and make a `pie-chart` of the species in the `tiny-sample`. What animals are in the sample? _____

- Click "Run" for a *new* random `tiny-sample`, and make *another* pie-chart for species. What animals are there? _____
- Click "Run" for a *new* random sample, and make *yet another* pie-chart for species. Based on these 3 samples, how many species do you think are at the shelter? _____
- Which is the *most common* species at the shelter? _____

3) What did you learn from taking multiple samples that you wouldn't have known if you'd only taken one?

_____

4) Repeat the steps above, but for `small-sample`. What animals are in the sample?

_____

5) Now that you've seen `small-sample`, how has your sense of the distribution of the species changed?

_____

6) Now use `medium-sample` to make a `pie-chart` of the species. If there are about 400 animals at the shelter, how many of each species would you predict there to be?

_____

7) Now use `large-sample` to make a `pie-chart` of the species. If there's anything you'd like to change about your prediction now that you've seen `large-sample`, record it here.

_____

8) Let's see how accurate your prediction is… *feel free to click "Run" and build a few more pie charts from your samples if you want to collect more information first!* When you're ready, make a `pie-chart` of `more-animals`.

- Which predictions were closest? _____
- Which predictions were off? _____
- Were there any surprises? _____

9) In the real world, we usually don't have access to a whole dataset to check predictions against! How could we test…

- *Every giraffe on the planet?*
- *Everyone who has ever come in contact with a covid-positive person?*
- *Every person who identifies as queer?*

What strategies can we use to make sure that predictions from samples are as close to accurate as possible?

_____

_____

Data Cycle: Measures of Center

Complete both of the Data Cycles shown here, which have questions defined to get you started.

Ask Questions

What is the mean age for cats at the shelter?

What question do you have?

What type of question is this? (circle one) Lookup Arithmetic Statistical

Consider Data

What Rows should we investigate? (All the rows, just the cats, fixed dogs, etc.)

What Columns do we need? (age, weight-in-kilograms, weeks, etc.)

Analyze Data

If you only need some rows, define your filter function here (Need help? Use the Design Recipe!)

If you need to make a new column, define your builder function here (Need help? Use the Design Recipe!)

What code that will make the table or display you want?

Interpret Data

What did you find out? What can you infer?

What new question(s) does this raise?

Ask Questions

What question do you have?

What type of question is this? (circle one) Lookup Arithmetic Statistical

Consider Data

What Rows should we investigate? (All the rows, just the cats, fixed dogs, etc.)

What Columns do we need? (age, weight-in-kilograms, weeks, etc.)

Analyze Data

If you only need some rows, define your filter function here (Need help? Use the Design Recipe!)

If you need to make a new column, define your builder function here (Need help? Use the Design Recipe!)

What code that will make the table or display you want?

Interpret Data

What did you find out? What can you infer?

What new question(s) does this raise?

Critiquing Written Findings

Consider the following dataset, representing the heaviest bench press (in lbs) for ten powerlifters:

```
135, 95, 230, 135, 203, 55, 1075, 135, 110, 185
```

1) In the space below, rewrite this dataset in sorted order.

2) In the table below, compute the measures of center for this dataset.

Mean (Average)	Median	Mode(s)

3) The following statements are correct … but misleading. Write down the reason why.

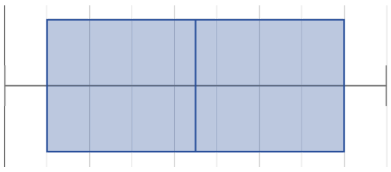
Statement	Why it's misleading
"More personal records are set at 135 lbs than any other weight!"	
"The average powerlifter can bench press 235 lbs."	
"With a median of 135, that means that half the people in this group can't even lift 135 lbs."	

Directions: Connect each item on this page to at least one other item by drawing an arrow and writing an explanation of how they are connected along the arrow. (Arrows may curve.)

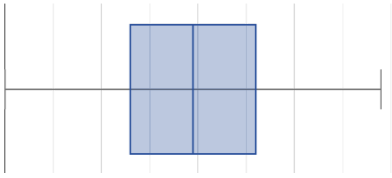
Minimum	Maximum	Quartile
Median	50%	Interquartile Range
Upper Quartile	Lower Quartile	25%

Matching Box Plots to Histograms

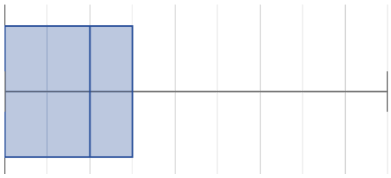
Match each box-plot to the histogram that displays the same data.



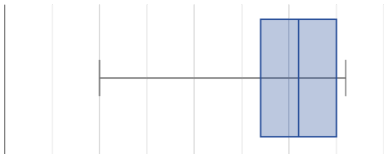
1



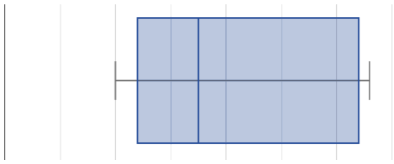
2



3

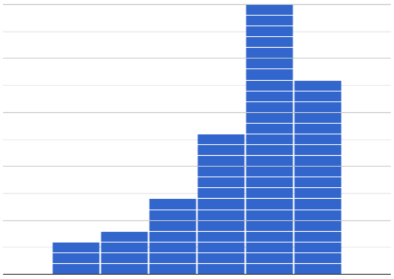


4

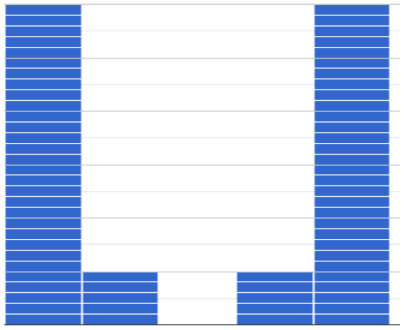


5

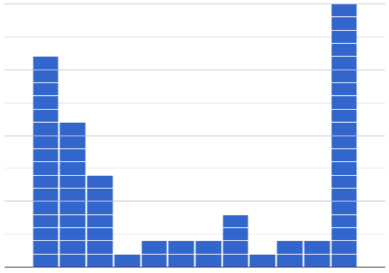
A



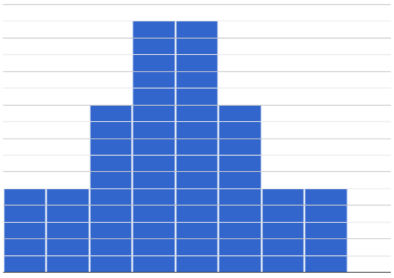
B



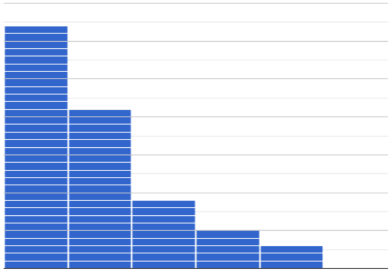
C



D



E



Computing Standard Deviation (2)

Shown below are ten different family incomes:

$43k, $62k, $39k, $141k, $58k, $82k, $41k, $73k, $68, $73k

1) Draw a **number line** representing the distribution of this data.

2) What is the mean income of the families in this dataset? _____

3) **Draw a star to locate the mean on the x-axis of the histogram above**.

4) For each family on the number line you drew, **draw a horizontal arrow** under the axis from your star to that family's interval, and **label the arrow with its distance from the mean**. (For example, if the mean is 50 and a family's income is $82k, your arrow would stretch from 50 to 82, and be labeled with the distance "32")

To compute the standard deviation we square each distance and take the average, then take the square root of the average.

5) For each of the 10 incomes in the table below, list the distance-from-mean for each income, using the mean you computed above. Then fill in the squared distance in the next row.

income (in 10s of thousand)	39	41	43	58	62	68	73	73	82	141
distance from mean										
squared distance										

6) Add all the squared distances. What is their sum? _____

7) What is the average squared distance? _____

8) Take the square root to find the **standard deviation**! _____

Matching Mean & Standard Deviation to Data

In the table below, match the mean and standard deviation to the list of data it describes.

Mean: 4 *StDev:* 0	**1**	**A**	`[list: -1,-2,-3,-4,-5,-6,-7]`
Mean: -4 *StDev:* 4	**2**	**B**	`[list: 1, 2, 3, 4, 5, 6, 7]`
Mean: 4 *StDev:* 2	**3**	**C**	`[list: -1, -9]`
Mean: 4 *StDev:* ~2.5	**3**	**D**	`[list: 0, 2, 3, 4, 5, 6, 8]`
Mean: -4 *StDev:* 2	**5**	**E**	`[list: 4, 4, 4, 4, 4]`

Mapping Examples with Circles of Evaluation

Contract: _____

Purpose Statement: _____

If I type…	→	It should map to…
EXAMPLE #1: Circle of Evaluation Code:	→	Circle of Evaluation: Code:
EXAMPLE #2: Circle of Evaluation Code:	→	Circle of Evaluation: Code:

Do The Examples Have the Same Contracts?

For each pair of Examples below, decide whether the two examples have the same contract. If they do, fill in the contract in the space provided. If not, write a few words explaining how you know their contracts aren't the same.

1) _____

```
examples:
  mystery(30) is 30 * 50
  mystery(10) is text("Welcome!", 10, "darkgreen")
end
```

2) _____

```
examples:
  mystery(30, 40) is 40 - (2 * 30)
  mystery(10, 15) is 15 - (2 * 10)
end
```

3) _____

```
examples:
  mystery("New York") is text("New York", 20, "red")
  mystery(20) is text("New York", 20, "red")
end
```

4) _____

```
examples:
  mystery("green", 32) is circle(32, "outline", "green")
  mystery(18, "green") is circle(18, "outline", "green")
end
```

5) _____

```
examples:
  mystery(6, 9, 10) is 6 / (9 + 10)
  mystery(3, 7) is 3 / (7 + 10)
end
```

6) _____

```
examples:
  mystery("red", "blue") is text("blue", 25, "red")
  mystery("purple", "Go Team!") is text("Go Team!", 25, "purple")
end
```

Do The Examples Have the Same Contracts? (2)

For each pair of Examples below, decide whether the two examples have the same contract. If they do, fill in the contract in the space provided. If not, write a few words explaining how you know their contracts aren't the same.

1) _____

```
examples:
  mystery(triangle(70, "solid", "green")) is
    triangle(140, "solid", "green")
  mystery(circle(100, "solid", "blue")) is
    circle(200, "solid", "blue")
end
```

2) _____

```
examples:
  mystery("red") is triangle(140, "solid", "red")
  mystery("blue", "circle") is circle(140, "solid", "blue")
end
```

3) _____

```
examples:
  mystery("+", 4, 5) is 4 + 5
  mystery("sqrt", 25) is num-sqrt(25)
end
```

4) _____

```
examples:
  mystery("circle", 4) is num-pi * num-sqr(4)
  mystery("square", 5) is num-sqr(5)
end
```

5) _____

```
examples:
  mystery("dog") is 3
  mystery("cat") is "kitten"
end
```

6) _____

```
examples:
  mystery("dog") is 3
  mystery("kitten") is 6
end
```

Matching Contracts and Examples

Match each Example on the left with its Contract on the right. NOTE: Multiple examples may match to the same contract!

Contract			Examples

`examples:`
`  match(circle(10, "solid", "green")) is`
`  rotate(37, circle(10, "solid", "green"))`
`end`
	1	A	`# match :: Number, Image -> Image`

`examples:`
`  match(triangle(20, "solid", "blue"), 3) is`
`  scale(3, triangle(20, "solid", "blue"))`
`end`
	2		

`examples:`
`  match(circle(20, "outline", "gold")) is`
`  rotate(37, circle(20, "outline", "gold"))`
`end`
	3	B	`# match :: Image, Number -> Image`

`examples:`
`  match(30, "red") is 30 + string-length("red")`
`end`
	4		

`examples:`
`  match(circle(10, "solid", "orange"), 22) is`
`  scale(22, circle(10, "solid", "orange"))`
`end`
	5		

`examples:`
`  match(10, "blue") is 10 + string-length("blue")`
`end`
	6	C	`# match :: Image -> Image`

`examples:`
`  match(5, star(20, "solid", "red")) is`
`  rotate(90 - 5, star(20, "solid", "red"))`
`end`
	7		

`examples:`
`  match(num-abs(-4), "45") is 4`
`end`
	8	D	`# match :: Number, String -> Number`

Matching Contracts and Examples 2

Match each Example on the left with its Contract on the right. NOTE: Multiple examples may match to the same contract!

Contract		Examples	

`examples:`
`  match(1.5) is "greater than 1"`
`end`
	1		

`examples:`
`  match(24) is star(24 * 2, "outline", "purple")`
`end`
	2		

`examples:`
`  match(string-length("tabletop")) is "8"`
`end`
	3	A	`# match :: Number -> String`

`examples:`
`  match(star(20, "outline", "red"), 3) is`
`  3 * image-height(star(20, "outline", "red"))`
`end`
	4	B	`# match :: Number -> Image`

`examples:`
`  match(circle(10, "solid", "silver"), 16) is`
`  16 * image-height(circle(10, "solid", "silver")`
`)`
`end`
	5	C	`# match :: Number, Number -> Number`

`examples:`
`  match("triangle", "blue") is triangle(40,`
`"outline", "blue")`
`end`
	6	D	`# match :: String, String -> Image`

`examples:`
`  match(30) is star(30 * 2, "outline", "purple")`
`end`
	7	E	`# match :: Images, Number -> Number`

`examples:`
`  match(string-length("coffee"), string-length(`
`"tea")) is`
`  6 + 3`
`end`
	8		

species-dot - Explore

Look at the code in the Custom Scatter Plot with Piecewise Functions Starter File.

What do you Notice?	What do you Wonder?

1) How is this program similar to the last one we looked at?

2) How is this code different from the program that made the `image-scatter-plot` with species labels?

3) Click "Run" and look at the display.

What do you Notice?	What do you Wonder?

Word Problem: species-dot

Directions: We want to generate a custom dot for our `image-scatter-plot`, such that every species gets a unique color. Write a function called `species-dot`, which takes in a Row from the animals table and returns a solid, 5px circle using black for dog, orange for cat, pink for rabbit, red for tarantula, green for lizard and blue for snail.

Every contract has three parts…

```
# species-dot:: Row -> Image
```

function name — *Domain* — *Range*

Write some examples, then circle and label what changes…

```
examples:
  species-dot(dog-row) is circle(5, "solid", "black")
  species-dot(cat-row) is circle(5, "solid", "orange")
  species-dot(rabbit-row) is circle(5, "solid", "pink")
  species-dot(tarantula-row) is circle(5, "solid", "red")
  species-dot(lizard-row) is circle(5, "solid", "green")
  species-dot(snail-row) is circle(5, "solid", "blue")
end
```

function name — *input(s)* — *what the function produces*

Write the definition, giving variable names to all your input values…

```
fun species-dot(r):
  if (r["species"] == "dog"): circle(5, "solid", "black")
  else if (r["species"] == "cat"): circle(5, "solid", "orange")
  else if (r["species"] == "rabbit"): circle(5, "solid", "pink")
  else if (r["species"] == "tarantula"): circle(5, "solid", "red")
  else if (r["species"] == "lizard"): circle(5, "solid", "green")
  else if (r["species"] == "snail"): circle(5, "solid", "blue")
  end
end
```

function name — *variable(s)*

Word Problem: sex-dot

Directions: We want to generate a custom dot for our `image-scatter-plot`, to distinguish female, hermaphrodite, and male animals. Write a function called `sex-dot`, which takes in a Row from the animals table and generates a 5-pixel solid dot that is red if the animal is female, purple if the animal is hermaphrodite and black if the animal is male.

Every contract has three parts…

_____ :: _____ -> _____

function name *Domain* *Range*

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____

function name *input(s)* *what the function produces*

_____ (_____) **is** _____

function name *input(s)* *what the function produces*

_____ (_____) **is** _____

function name *input(s)* *what the function produces*

end

Write the definition, giving variable names to all your input values…

fun _____ (_____):

function name *variable(s)*

_____

what the function does with those variable(s)

_____

what the function does with those variable(s)

_____

what the function does with those variable(s)

end

Animal-Image - Explore

Open the Custom Animals Starter File and look at the code.

What do you notice?	What do you wonder?

1) How is this program similar to the last one we looked at?

2) How is the code different from the other `image-scatter-plot` programs we've seen?

3) Click "Run". How does using clip art help us to better understand the data?

4) What risks might there be to using clip art for data that describes humans?

5) We have seen `image-scatter-plot` used to make a lot of different custom scatter plots. How could you use `image-scatter-plot` in your own analysis?

Word Problem: fixed-dot

Directions: We want to generate a custom dot for our `image-scatter-plot`, to distinguish between animals that are fixed and unfixed. Write a function called `fixed-dot`, which takes in a Row from the animals table and generates a 5-pixel solid green dot if the animal is fixed and a 5-pixel black dot if the animal is not fixed.

Every contract has three parts…

```
# _____ :: _____ -> _____
```

function name — *Domain* — *Range*

Write some examples, then circle and label what changes…

```
examples:
  _____ ( _____ ) is _____
  _____ ( _____ ) is _____
end
```

function name — *input(s)* — *what the function produces*

Write the definition, giving variable names to all your input values…

```
fun _____ ( _____ ):
  _____
  _____

end
```

function name — *variable(s)* — *what the function does with those variable(s)* — *what the function does with those variable(s)*

Dots for Value Ranges - Explore

Open the Dots for Value Ranges Starter File and look at the code.

What do you notice?	What do you wonder?

1) How is this program similar to the last one we looked at?

2) How is the code for `weight-dot` different from other functions we've seen?

3) Can you explain how this code works?

What Table Do We Get?

You have the following functions defined below:

```
fun lookup-fixed(animal): animal["fixed"]                end
fun is-dog(animal):       animal["species"] == "dog"     end
fun is-old(animal):       animal["age"] > 10             end
fun label(animal):        text(animal["name"], 20, "red") end
```

The table `t` below represents four animals from the shelter:

name	sex	age	fixed	species	pounds
"Toggle"	"female"	12	true	"dog"	48
"Fritz"	"male"	4	false	"dog"	92
"Nori"	"female"	6	true	"dog"	35.3
"Sunflower"	"female"	2	false	cat	51.6

Below is a list of expressions that use table methods. *Match* each expression to the description of the table we should get after evaluating each one.

`t.order-by("age", true)`	1	A	Produces a table with Toggle, Fritz, and Nori - but not Sunflower.
`t.order-by("pounds", false)`	2	B	Produces a table of all four animals, sorted youngest-to-oldest
`t.build-column("sticker", label)`	3	C	Produces a table, with only Toggle.
`t.filter(is-old)`	4	D	Produces an identical table with an extra column called "dog", whose values are true, true, true, false
`t.filter(lookup-fixed)`	5	E	Produces a table containing only Nori and Toggle.
`t.filter(is-dog)`	6	F	Produces a table with all four animals, sorted from heaviest to lightest.
`t.build-column("dog", is-dog)`	7	G	Won't run: will produce an error
`t.filter(label)`	8	H	Produces an identical table with an extra column called "sticker", whose values are images

Writing Examples from Purpose Statements 2

We've provided contracts and purpose statements to describe two different functions. Write examples for each of those functions.

Contract and Purpose Statement

Every contract has three parts…

half-image:: *String* -> *Image*

function name *Domain* *Range*

Consumes an image, and produces that image scaled to half its size.

what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is**

function name *input(s)*

_____

what the function produces

_____ (_____) **is**

function name *input(s)*

_____

what the function produces

end

Contract and Purpose Statement

Every contract has three parts…

product-squared:: *Number, Number* -> *Number*

function name *Domain* *Range*

Consumes two numbers and squares their product

what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____

function name *input(s)* *what the function produces*

_____ (_____) **is** _____

function name *input(s)* *what the function produces*

end

Rocket Height Challenges

1) Can you make the rocket fly faster?

2) Can you make the rocket fly slower?

3) Can you make the rocket sink down instead of fly up?

4) Can you make the rocket accelerate over time, so that it moves faster the longer it flies?

5) Can you make the rocket blast off and then land again?

6) Can you make the rocket blast off, reach a maximum height of exactly 1000 meters, and then land?

7) Can you make the rocket blast off, reach a maximum height of exactly 1000 meters, and then land after exactly 100 seconds?

8) Can you make the rocket fly to the edge of the the universe?

Blank Design Recipe

Directions:

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

Directions:

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

The Design Recipe (Restaurants)

Directions: Use the Design Recipe to write a function `split-tab` that takes in a cost and the number of people sharing the bill and splits the cost equally.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

Directions: Use the Design Recipe to write a function `tip-calculator` that takes in the cost of a meal and returns the 15% tip for that meal.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

The Design Recipe (Direct Variation)

Directions: Use the Design Recipe to write a function `wage`, that takes in a number of hours worked and returns the amount a worker will get paid if their rate is $10.25/hr.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

Directions: On average, people burn about 11 calories/minute riding a bike. Use the Design Recipe to write a function `calories-burned` that takes in the number of minutes you bike and returns the number of calories burned. .

Contract and Purpose Statement

Every contract has three parts…

_____ :: *Number* -> *Number*
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

The Design Recipe (Slope/Intercept 1)

Directions: For his birthday, James' family decided to open a savings account for him. He started with $50 and committed to adding $10 a week from his afterschool job teaching basketball to kindergartners. Use the Design Recipe to write a function `savings` that takes in the number of weeks since his birthday and calculates how much money he has saved.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

Directions: Use the Design Recipe to write a function `moving` that takes in the days and number of miles driven and returns the cost of renting a truck. The truck is $45 per day and each driven mile is 15¢.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

The Design Recipe (Negative Slope/Intercept)

Directions: An Olympic pool holds 660,000 gallons of water. A fire hose can spray about 250 gallons per minute. Use the Design Recipe to write a function `pool` that takes in the number of minutes that have passed and calculates how much water is still needed to fill it.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

Directions: The community arts fund awards a $1500 grant each month to support a new mural. They started with $50000 in their account. Use the Design Recipe to write a function `funds-available` that takes in the number of months and calculates how much money they have left.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

The Design Recipe (Geometry - Rectangles)

Directions: Use the Design Recipe to write a function `lawn-area` that takes in the length and width of a rectangular lawn and returns its area.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

Directions: Use the Design Recipe to write a function `rect-perimeter` that takes in the length and width of a rectangle and returns the perimeter of that rectangle.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

The Design Recipe (Geometry - Rectangular Prisms)

Directions: Use the Design Recipe to write a function `rectprism-vol` that takes in the length, width, and height of a rectangular prism and returns the Volume of a rectangular prism.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

Directions: Use the Design Recipe to write a function `rect-prism-sa` that takes in the width, length and height of a rectangular prism and calculates its surface area (the sum of the areas of each of its six faces)

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is**
function name *input(s)*

_____
what the function produces

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

The Design Recipe (Geometry - Circles)

Directions: Use the Design Recipe to write a function `circle-area-dec` that takes in a radius and uses the decimal approximation of pi (3.14) to return the area of the circle.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

Directions: Use the Design Recipe to write a function `circumference` that takes in a radius and uses the decimal approximation of pi (3.14) to return the circumference of the circle.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

The Design Recipe (Geometry - Cylinders)

Directions: Use the Design Recipe to write a function `circle-area` that takes in a radius and uses the fraction approximation of pi ($\frac{22}{7}$) to return the area of the circle.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

Directions: Use the Design Recipe to write a function `cylinder` that takes in a cylinder's radius and height and calculates its volume, making use of the function *circle-area*.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

The Design Recipe (Breaking Even)

Directions: The Swamp in the City Festival is ordering t-shirts. The production cost is $75 to set up the silk screen and $9 per shirt. Use the Design Recipe to write a function `min-shirt-price` that takes in the number of shirts to be ordered, *n*, and returns the minimum amount the festival should charge for the shirts in order to break even. (Assume that they will sell all of the shirts.)

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

The Design Recipe (Marquee & Cubing)

Directions: Use the Design Recipe to write a function `marquee` that takes in a message and returns that message in large gold letters.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

Directions: Use the Design Recipe to write a function `num-cube` that takes in a number and returns the cube of that number.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

Design Recipe Telephone Set 2: is-old

Directions: Define a function called `is-old`, which consumes a Row of the animals table and *computes* whether it is more than 10 years old.
HINT: use predefined rows like `old-row` to make your examples easier!

Contract and Purpose Statement

Every contract has three parts…

is-old:: *Row* -> *Boolean*

function name *Domain* *Range*

_____

what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____

function name *input(s)* *what the function produces*

_____ (_____) **is** _____

function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):

function name *variable(s)*

_____

what the function does with those variable(s)

end

_____

Design Recipe Telephone Set 2: kilos

Directions: Define a function called `kilos`, which consumes a Row of the animals table and computes the number of kilograms the animal weighs (1 kg = 2.2lbs). HINT: use predefined rows like `cat-row` to make your examples easier!

Contract and Purpose Statement

Every contract has three parts…

kilos:: *Row* -> *String*

function name *Domain* *Range*

_____

what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____

function name *input(s)* *what the function produces*

_____ (_____) **is** _____

function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):

function name *variable(s)*

_____

what the function does with those variable(s)

end

_____

Design Recipe Telephone Set 2: is-cat

Directions: Define a function called `is-cat`, which consumes a `Row` of the animals table and *computes* whether the animal is a cat. HINT: use predefined rows like `cat-row` to make your examples easier!

Contract and Purpose Statement

Every contract has three parts…

is-cat:: *Row* -> *Boolean*

function name *Domain* *Range*

_____

what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____

function name *input(s)* *what the function produces*

_____ (_____) **is** _____

function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):

function name *variable(s)*

_____

what the function does with those variable(s)

end

_____

Design Recipe

Directions:

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

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what the function does with those variable(s)

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function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

Data Cycle: Analyzing Dogs

Ask Questions

How many young dogs are fixed v. unfixed?

What question do you have?

What type of question is this? (circle one) Lookup Arithmetic Statistical

Consider Data

What Rows should we investigate? (All the rows, just the cats, fixed dogs, etc.)

What Columns do we need? (age, weight-in-kilograms, weeks, etc.)

Analyze Data

If you only need some rows, define your filter function here (Need help? Use the Design Recipe!)

If you need to make a new column, define your builder function here (Need help? Use the Design Recipe!)

What code that will make the table or display you want?

Interpret Data

What did you find out? What can you infer?

What new question(s) does this raise?

Ask Questions

Is there a relationship between weight and days-to-adoption for dogs?

What question do you have?

What type of question is this? (circle one) Lookup Arithmetic Statistical

Consider Data

What Rows should we investigate? (All the rows, just the cats, fixed dogs, etc.)

What Columns do we need? (age, weight-in-kilograms, weeks, etc.)

Analyze Data

If you only need some rows, define your filter function here (Need help? Use the Design Recipe!)

If you need to make a new column, define your builder function here (Need help? Use the Design Recipe!)

What code that will make the table or display you want?

Interpret Data

What did you find out? What can you infer?

What new question(s) does this raise?

Data Cycle: Analyzing Cats

Ask Questions

Are more old cats are fixed than unfixed unfixed?

What question do you have?

What type of question is this? (circle one) Lookup Arithmetic Statistical

Consider Data

What Rows should we investigate? (All the rows, just the cats, fixed dogs, etc.)

What Columns do we need? (age, weight-in-kilograms, weeks, etc.)

Analyze Data

If you only need some rows, define your filter function here (Need help? Use the Design Recipe!)

If you need to make a new column, define your builder function here (Need help? Use the Design Recipe!)

What code that will make the table or display you want?

Interpret Data

What did you find out? What can you infer?

What new question(s) does this raise?

Ask Questions

Is there a relationship between the number of kilograms that a cat weighs, and how long it takes to be adopted?

What question do you have?

What type of question is this? (circle one) Lookup Arithmetic Statistical

Consider Data

What Rows should we investigate? (All the rows, just the cats, fixed dogs, etc.)

What Columns do we need? (age, weight-in-kilograms, weeks, etc.)

Analyze Data

If you only need some rows, define your filter function here (Need help? Use the Design Recipe!)

If you need to make a new column, define your builder function here (Need help? Use the Design Recipe!)

What code that will make the table or display you want?

Interpret Data

What did you find out? What can you infer?

What new question(s) does this raise?

The Design Recipe

Write helper functions for **your** dataset, which you can use to define grouped samples. Since all helper functions will consume Rows, their Domains have already been filled in for you.

Directions: Define a function called _____, which consumes a Row of the _____ table and _____.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ (r :: Row) _____ -> Boolean

function name *Domain* *Range*

_____

what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____

function name *input(s)* *what the function produces*

_____ (_____) **is** _____

function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):

function name *variable(s)*

_____

what the function does with those variable(s)

end

Directions: Define a function called _____, which consumes a Row of the _____ table and _____.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ (r :: Row) _____ -> Boolean

function name *Domain* *Range*

_____

what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____

function name *input(s)* *what the function produces*

_____ (_____) **is** _____

function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):

function name *variable(s)*

_____

what the function does with those variable(s)

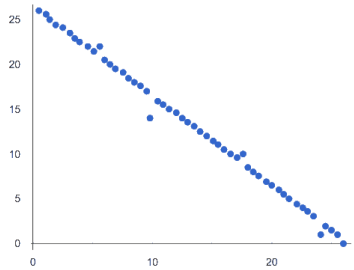
end

Identifying Form, Direction and Strength (Matching)

Match the description (left) with the scatter plot (right).

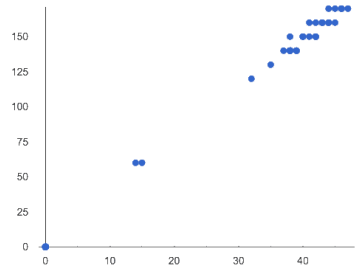
This data has a moderate negative correlation. **1**

A



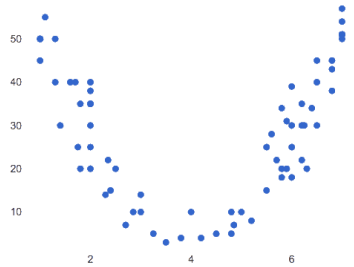
This dataset is non-linear. **2**

B



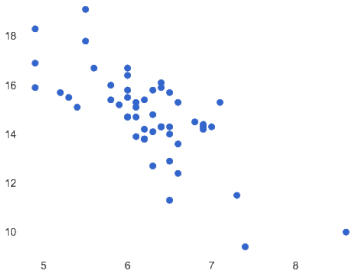
There is no correlation in this data. **3**

C



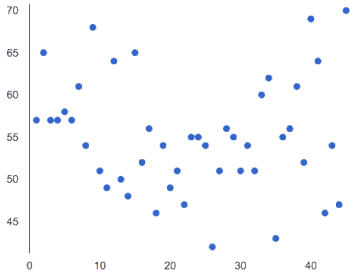
There is a strong positive correlation in this data. **4**

D



There is a strong negative correlation in this data. **5**

E



Age vs. Height Explore

Open the Age vs. Height Starter File and click "Run" to interact with data from another sample of students.

1) Take a look at the code in the Definitions Area. What do you notice? What do you wonder?

2) Build `image-scatter-plot(h-table, "age", "height", dot)`. Try to visualize the line of best fit for just the blue dots. Then try to visualize the line of best fit for just the red stars. How do you think they would compare? Which line do you think would be steeper?

3) Make three linear regression plots comparing `age` and `height`, and record the results for each in the table below:

- The whole population: `lr-plot(h-table, "gender-id", "age", "height")`
- Females only: `lr-plot(h-table.filter(is-f), "gender-id", "age", "height")`
- Males only: `lr-plot(h-table.filter(is-m), "gender-id", "age", "height")`

Sample	rate of change	y-intercept	R value
All			
Females			
Males			

4) What makes it difficult to compare these plots visually?

Rebuild `lr-plot(h-table.filter(is-f), "gender-id", "age", "height")`, adjust the window of the interactive plot so that:

x-min:	x-max:	y-min:	y-max:
6.5	12.5	45	70

and click `Redraw`. Then, do the same for `lr-plot(h-table.filter(is-m), "gender-id", "age", "height")`.

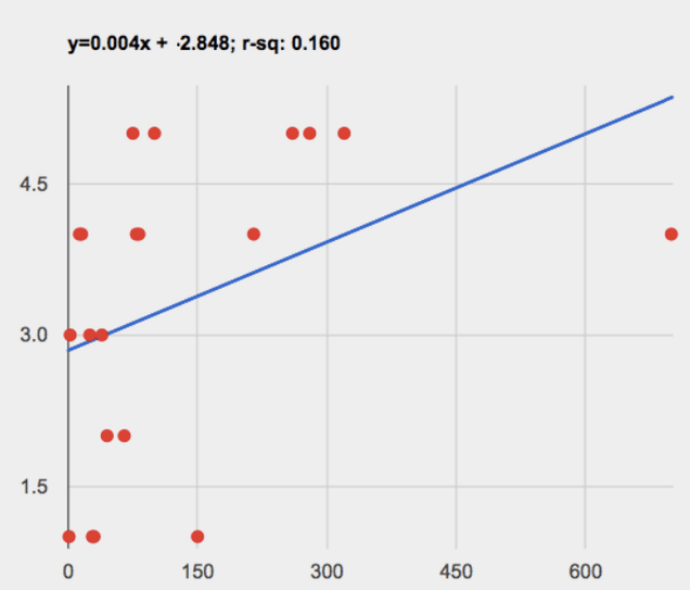
5) How do the plots compare now that their windows match?

6) What happens if you compare the students' height in inches to their height in centimeters by plotting `lr-plot(h-table, "gender-id", "height-cm", "height")`?

Describing Relationships (2)

Are people more satisfied with a purchase on which they spent more money? A group of people were surveyed about their most recent purchase, rating their satisfaction from 1 (very unsatisfied) to 5 (extremely satisfied).

Dollars	Satisfaction
15.5	4
280	5
0.99	1
2.3	3
39	3
82	4
215	4
700	4
25	3
79	4
99.99	5
30	1
75	5
13	4
320	5
260	5
150	1
28	1
45	2
65	2



Describe the relationship between dollars spent and satisfaction shown in the data above.

Data Cycle: Regression Analysis 2

Ask Questions	_____ *What question do you have?* _____ What type of question is this? (circle one) Lookup Arithmetic Statistical
Consider Data	_____ *What Rows should we investigate? (All the rows, just the cats, fixed dogs, etc.)* _____ *What Columns do we need? (age, weight-in-kilograms, weeks, etc.)*
Analyze Data	_____ *If you only need some rows, define your filter function here (Need help? Use the Design Recipe!)* _____ *If you need to make a new column, define your builder function here (Need help? Use the Design Recipe!)* _____ *What code that will make the table or display you want?*
Interpret Data	_____ *What did you find out? What can you infer?* _____ _____ *What new question(s) does this raise?* _____ _____

Write your Data Story below:

I performed a linear regression on a sample of _____ (dataset or subset) and found _____ (a weak/strong/moderate (R=...), positive/negative) correlation between _____ ([x-axis]) and _____ ([y-axis]). I would predict that a 1 _____ ([x-axis units]) increase in _____ ([x-axis]) is associated with a _____ ([slope, y-units]) _____ ([increase/decrease]) in _____ ([y-axis]).

Selection Bias or Biased Study?

The school newspaper ran an article stating that chicken was more popular than pork in the East Village.

Kendell thinks the study was biased.

> *Would you rather eat pork or delicious crispy fried chicken?* That's such a leading question! It encouraged people to pick chicken. I bet the results would have been differet if they had asked about crispy bacon!

Carson thinks the study suffered from selection bias.

> One of the survey sites was outside of a mosque?! Muslims don't even eat pork!

Who's right? How do you know?

Identifying Threats to Validity (Part 1)

Data scientists surveyed **100 coffee and tea drinkers over a 1-month period** to determine a connection between caffeine consumption and sleep disorders. Respondents were asked questions regarding the **amount of coffee or tea consumed daily, dietary habits, sleep habits, and exercise levels**. Based on this survey, the data scientists claim that **people who drink tea exclusively have moreincidents of insomnia than people who drank coffee exclusively**.

Before publishing their findings, the data scientists have asked you to review their claim. In the space below, indicate possible **threats to validity** faced by this study.

Identifying Threats to Validity (Part 2)

Data scientists conducted a study of **15,000 people living in Alaska and New Mexico**. They wanted to find out if people in colder climates spent more money on footwear. They discovered that **Alaskans spent an average of $150 per year on shoes, whereas New Mexicans spent an average of $45 per year**.

Before publishing their findings, the data scientists have asked you to review their claim. In the space below, indicate possible **threats to validity** faced by this study.

Identifying Threats to Validity (Part 3)

Data scientists want to know if listening to music or podcasts reduce symptoms of stress in individuals. They conducted **a study of 1,000 people who were brought into a laboratory office for testing**. While wearing a heart-rate monitor, participants were asked to listen to either music or a podcast of their choosing while completing a series of complicated puzzles. The data scientists discovered that on average, **participants who listened to music had a 5% lower heart rate while completing the tasks than those who listened topodcasts**.

Before publishing their findings, the data scientists have asked you to review their claim. In the space below, indicate possible **threats to validity** faced by this study.